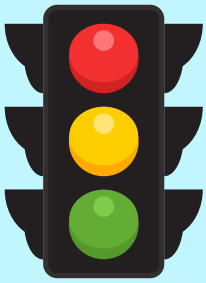


Activity 1



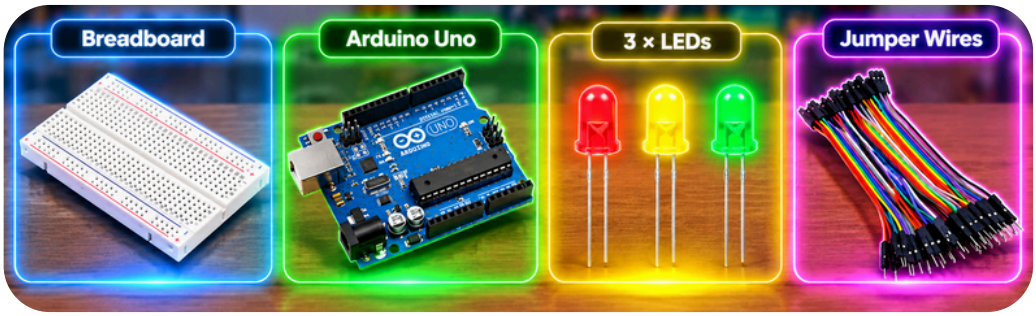
Activity Name: Traffic Light Controller



The Problem: Imagine your school is having a big fair, and many cars and people are moving near the entrance. Without traffic signals, cars may stop at the wrong time, causing confusion and making it difficult for everyone to move safely. The students notice that a simple system is needed to tell vehicles when to stop, wait, and go. How can we create a small working model that controls traffic in the right sequence?

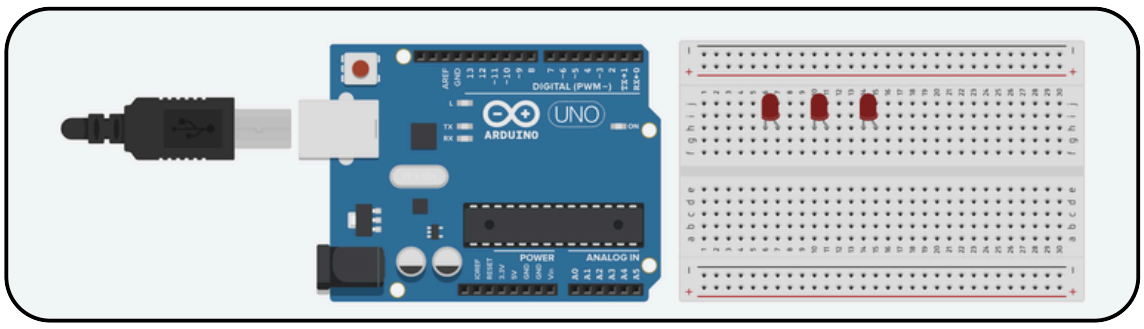
What Will We Build: The students will be able to build a mini traffic light system using three LEDs and an Arduino. The LEDs will turn Red, Yellow, and Green in sequence, just like a real traffic signal.

Components Required

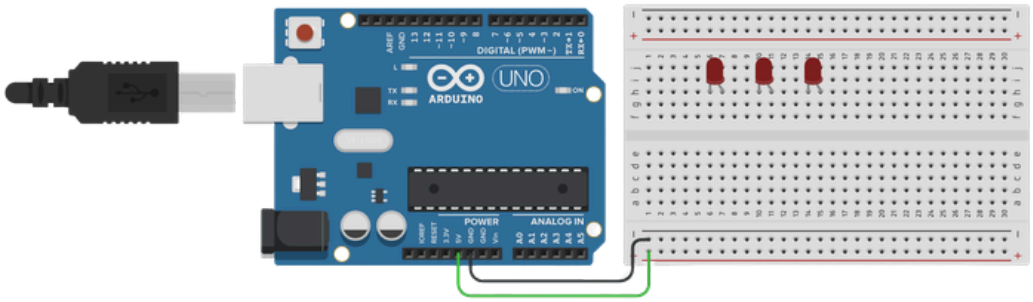


Circuit Designing

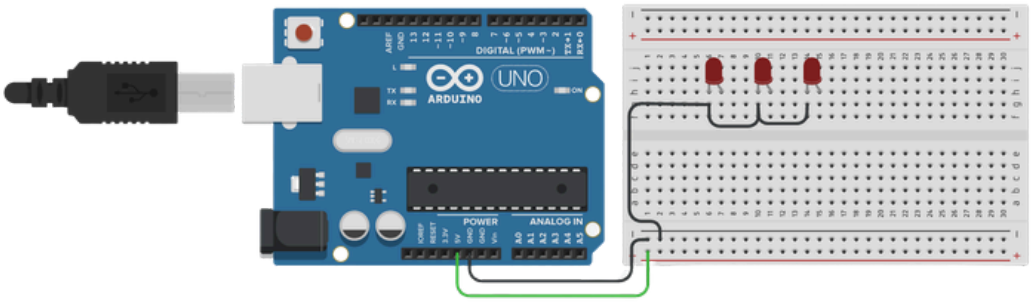
Step 1: Connect all the LEDs to the breadboard



Step 2: Connect the positive and negative terminals of the breadboard to 5V and GND pin of the Arduino Uno.

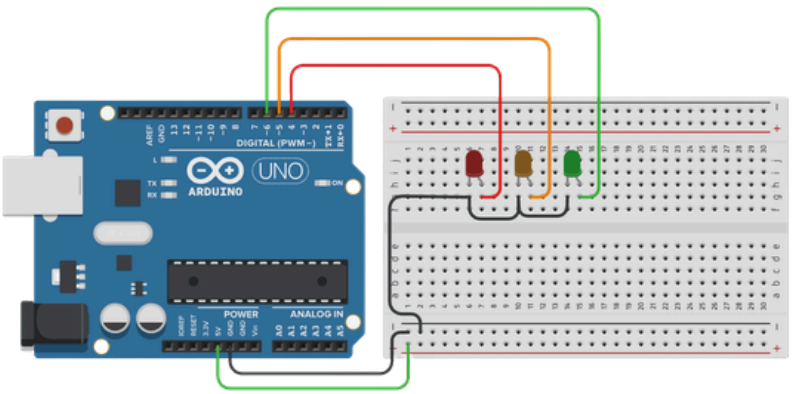


Step 3: Connect all the negative terminals of all LEDs to the negative pin of the breadboard.



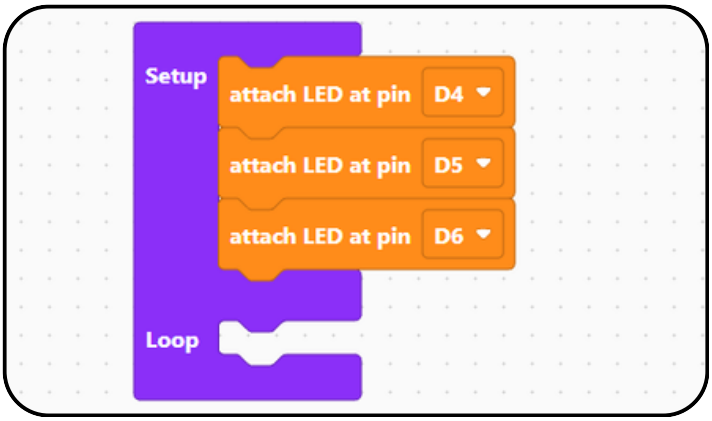
Step 4: Connect the LEDs' anode pins to the Arduino Uno.

LED Pins	Arduino Pins
Red LED	Digital Pin 4
Yellow LED	Digital Pin 5
Green LED	Digital Pin 6

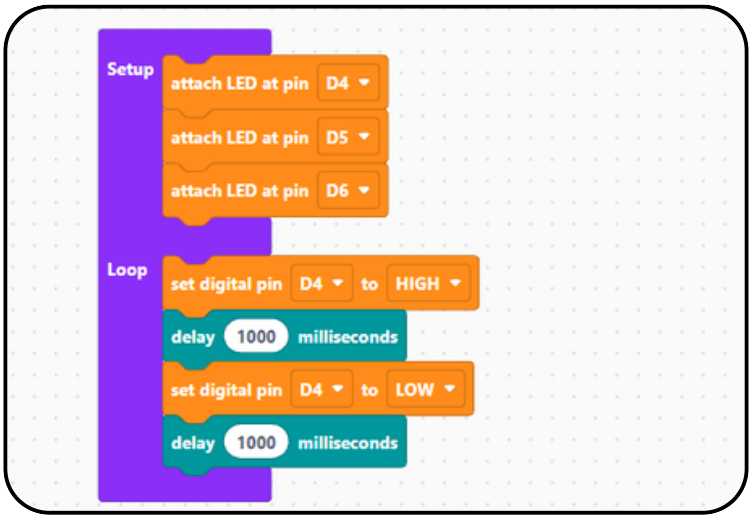


Project Coding

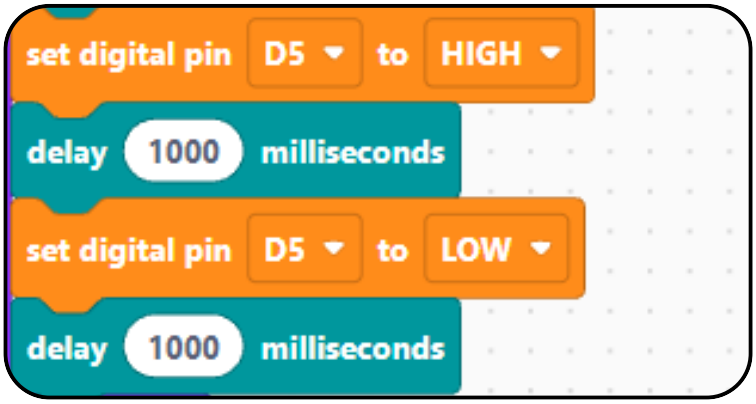
Step 1: Make code to intialize the pins.



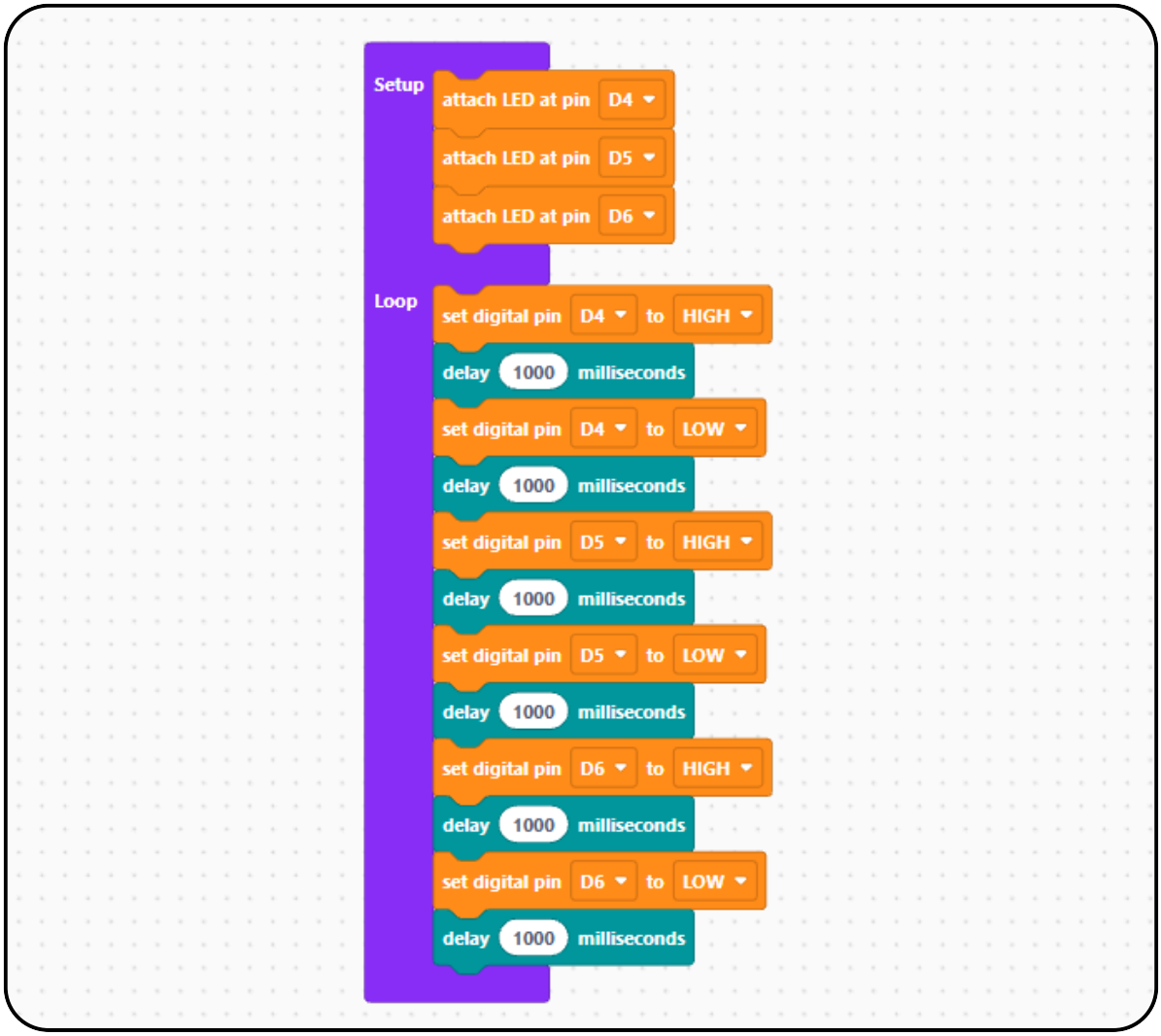
Step 2: Make code to turn ON the RED LED.



Step 3: Make code to turn ON the Yellow LED



Step 4: Make code to turn ON the Green LED



Learning Outcomes

- The students will be able to identify the main parts used to make a traffic light model.
- The students will be able to connect three LEDs to an Arduino using a breadboard.
- The students will be able to create simple code to control the Red, Yellow, and Green LEDs.



BRAIN TEASERS

Section A – Multiple Choice Questions

1. Which board is used to control the LEDs?

A. Arduino Uno
B. Keyboard
C. Speaker
D. Mouse

2. How many LEDs are used in the traffic light project?

A. 1
B. 2
C. 3
D. 5

3. Which LED represents “Stop” in a traffic light?

A. Green
B. Yellow
C. Blue
D. Red

Section B – True or False

1. A breadboard is used to connect the LEDs in the project.....

2. The project uses only one LED.....

3. The LEDs can be programmed to turn ON one after another.....